continuing to exist in a changing environment is to manage correlations with the external world better, that is to say, information; to collect, store, transmit and elaborate information. For this reason, DNA exists, together with immune systems, sense organs, nervous systems, complex brains, languages, books, the library of Alexandria, computers and Wikipedia: they maximize the efficiency of information management – the management of correlations favouring survival.

The statue that Aristotle sees in a block of marble is more than the block of marble: but it is not an abstract form that resides just in the statue. It is something residing in the correlations between the mind of Aristotle, or ours, and the marble; something that pertains to the information which the marble provides regarding something that is significant for Aristotle, or for us. It is something regarding a discus thrower, Phidias, Aristotle and the marble, and resides in the correlated dispositions of the atoms of the statue, and the correlations between these and a thousand others, in our minds or in Aristotle's. These speak of a discus thrower, just as the white ball in your hand tells you that the ball in mine is black. We are structured to manage precisely this – information – and remain in existence thanks to this.

Even from this brief overview it should be clear that the notion of information plays a central role in our attempts to understand the world. From communication to the basis of genetics, from thermodynamics to quantum mechanics and up to quantum gravity, the notion of information is gaining ground as a tool for understanding. The world should not be understood as an amorphous ensemble of atoms – but rather as a game of mirrors, founded on the correlations between the structures formed by *combinations* of these atoms.

As Democritus said, it is not just a question of these atoms but also of the *order* in which they are arranged. Atoms are like the letters in an alphabet: an extraordinary alphabet, so rich as to be able to read, reflect and even think about itself. We are not atoms; we are *orders* in which atoms are arranged, capable of mirroring other atoms and mirroring ourselves.

Democritus gave a strange definition of 'man': 'Man is what we all know.'[2] At first sight, this seems rather silly and empty, but it is not so.

Salomon Luria, the major scholar of Democritus, observes that it is not a banality that Democritus is giving us. The nature of a man is not his internal structure but the network of personal, familial and social interactions within which he exists. It is these which 'make' us, these which guard us. As humans, we are that which others know of us, that which we know of ourselves, and that which others know about our knowledge. We are complex nodes in a rich web of reciprocal information.

All of this is not yet a theory. These are tracks we are following, I believe, in seeking to understand the world around us better. There still remains a